

Task 1: Install Apache and PHP

Install Apache and PHP

- `sudo dnf install httpd php php-cli -y`

Enable and Start Apache

- `sudo systemctl enable --now httpd`

Verify Status

- `sudo systemctl status httpd`

Allow HTTP Through Firewall

- `sudo firewall-cmd --permanent --add-service=http`
- `sudo firewall-cmd --reload`

Create index.php

- `sudo vi /var/www/html/index.php`

Add:

```
<?php
echo "<h1>Welcome to My Foodie Blog</h1>";
?>
```

Restart Apache

- `sudo systemctl restart httpd`

Test:

- `http://<server-ip>/index.php`

Task 2: Create Custom 404 Error Page

Create 404 Page

- `sudo vi /var/www/html/404.html`

Add:

```
<!DOCTYPE html>
<html>
<head>
<title>404</title>
</head>
<body>
<h1>Oops! This page is as empty as your cart.</h1>
```

```
</body>
</html>
```

Configure Apache

- `sudo vi /etc/httpd/conf/httpd.conf`

Add near the bottom:

- `ErrorDocument 404 /404.html`

Restart Apache

- `sudo systemctl restart httpd`

Test

- `http://<server-ip>/xyz`

Task 3: Install FTP Server (vsftpd)

Install

- `sudo dnf install vsftpd -y`

Enable Service

- `sudo systemctl enable --now vsftpd`

Verify

- `sudo systemctl status vsftpd`

Open Firewall

- `sudo firewall-cmd --permanent --add-service=ftp`
- `sudo firewall-cmd --reload`

Upload Sample Image

copy directly

- `sudo cp sample.jpg /var/www/html/`

Verify

- `http://<server-ip>/sample.jpg\`

Task 4: Configure Node.js Service

Install Node.js

- `sudo dnf module install nodejs -y`

Create folder

- `mkdir ~/playlist`

- cd ~/playlist

Create file

- vi app.js

Paste

```
const http = require('http');
http.createServer((req,res)=>{
  res.writeHead(200,{'Content-Type':'application/json'});
  res.end(JSON.stringify({
    playlist:[
      "Perfect",
      "Believer",
      "Closer",
      "Shape of You"
    ]
  }));
}).listen(3000);
console.log("Server running on port 3000");
```

Run

- node app.js

Open Firewall

- sudo firewall-cmd --permanent --add-port=3000/tcp
- sudo firewall-cmd --reload

Test

- http://<server-ip>:3000

Task 5: Restrict Access to /secret

Create Directory

- sudo mkdir /var/www/html/secret

Create Page

- sudo vi /var/www/html/secret/index.html

Add

- <h1>Secret Area</h1>

Configure Apache

Create configuration

- `sudo vi /etc/httpd/conf.d/secret.conf`

Add

```
<Directory "/var/www/html/secret">  
Require ip 127.0.0.1  
</Directory>
```

Restart Apache

- `sudo systemctl restart httpd`

Test

From localhost

- `http://127.0.0.1/secret`

From another computer

- `http://<server-ip>/secret`